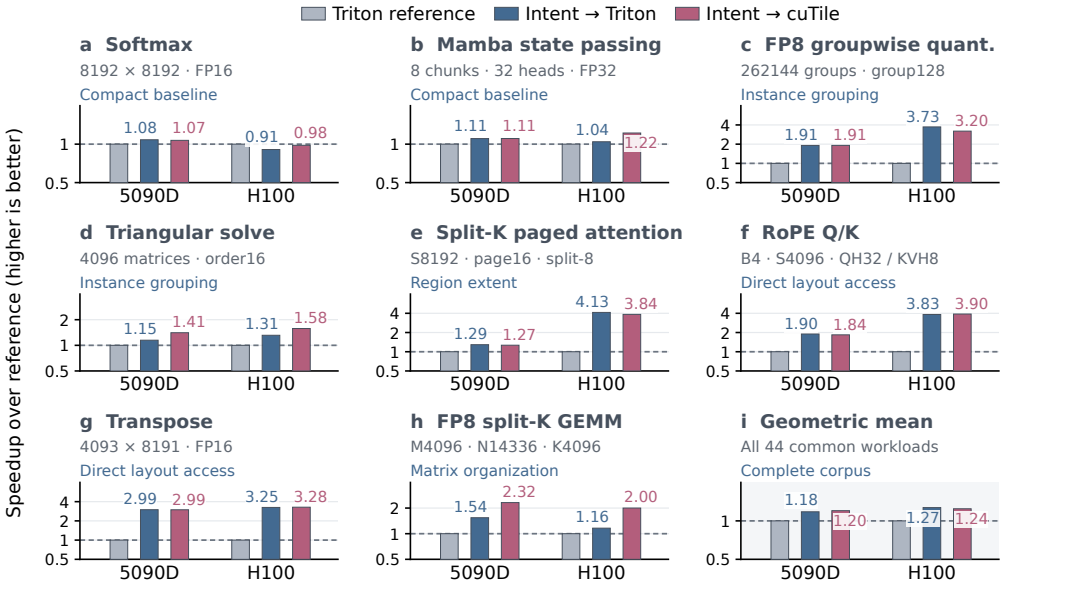
Triton reference
Intent → Triton
Intent → cuTile
Speedup over reference (higher is better)
a Softmax
8192 × 8192 · FP16
Compact baseline
1.08
1.07
0.91
0.98
b Mamba state passing
8 chunks · 32 heads · FP32
Compact baseline
1.11
1.11
1.04
1.22
c FP8 groupwise quant.
262144 groups · group128
Instance grouping
1.91
1.91
3.73
3.20
d Triangular solve
4096 matrices · order16
Instance grouping
1.15
1.41
1.31
1.58
e Split-K paged attention
S8192 · page16 · split-8
Region extent
1.29
1.27
4.13
3.84
f RoPE Q/K
B4 · S4096 · QH32 / KVH8
Direct layout access
1.90
1.84
3.83
3.90
g Transpose
4093 × 8191 · FP16
Direct layout access
2.99
2.99
3.25
3.28
h FP8 split-K GEMM
M4096 · N14336 · K4096
Matrix organization
1.54
2.32
1.16
2.00
i Geometric mean
All 44 common workloads
Complete corpus
1.18
1.20
1.27
1.24
5090D
H100